Reactor Maintenance Handbook

Lakeside Research Reactor · Revision 7 · Approved for internal use

1. Purpose

This handbook sets out the routine maintenance programme for systems that support safe operation of the Lakeside research reactor. It describes inspection intervals, the records that must be kept, and the conditions that require maintenance outside the normal schedule.

The handbook applies to all maintenance staff, contractors working under a facility work permit, and supervisors who approve completed work. Where this handbook and a vendor manual disagree, the more conservative requirement applies until the discrepancy is resolved by the maintenance engineer.

2. How to use this handbook

Each system chapter lists scheduled tasks first, followed by condition-based tasks. Intervals are measured in days of normal operation unless the chapter states otherwise. Time spent in an extended shutdown does not count toward an interval.

3. Records and sign-off

Every maintenance task produces a record in the facility maintenance log. The record identifies the equipment, the task performed, the readings taken, any parts replaced, and the name of the person who completed the work.

Completed work is reviewed by a supervisor within five working days. Records with missing readings are returned to the technician and the task is treated as incomplete until the readings are supplied.

4. Planning the annual outage

Tasks that require the reactor to be shut down are grouped into a planned outage. The outage plan is issued at least eight weeks in advance so that parts, permits, and contractor support can be arranged.

5. Primary coolant system

The primary coolant system removes heat from the core and carries it to the heat exchanger. Its pumps, valves, and instrumentation are safety-related and are maintained to the schedule in this chapter.

5.1 Coolant pumps: scheduled checks

Primary coolant pumps shall receive a visual and vibration check every 30 days of normal operation. Where vibration exceeds the recorded baseline by more than 20 percent, an inspection shall be performed within 72 hours.

Vibration readings are taken at the inboard and outboard bearings using the handheld analyser kept in the control room. Readings are entered in the pump log on the same shift.

5.2 Coolant pumps: mechanical inspection

A full mechanical inspection of each primary coolant pump, including seals and bearings, shall be completed every 18 months and scheduled to coincide with the planned outage.

The inspection follows the vendor procedure and includes seal leakage measurement, bearing clearance checks, and coupling alignment. Replacement seals are installed whenever leakage exceeds the vendor limit.

5.3 Valves and instrumentation

Isolation valves are stroke-tested quarterly. Flow and temperature instruments are calibrated annually against reference instruments with current calibration certificates.

6. Secondary systems

Secondary cooling, ventilation, and electrical supply systems are maintained according to their vendor schedules unless this handbook sets a shorter interval. Ventilation filters are inspected monthly and replaced when the pressure drop reaches the marked limit.

7. Condition-based maintenance

Any technician who observes abnormal noise, leakage, or temperature should report it to the shift supervisor immediately. The supervisor decides whether the equipment may remain in service until the next scheduled check.

8. Revision history

Revision 7 shortened the vibration response time from five days to 72 hours and added the requirement to log readings on the same shift. Revision 6 introduced the combined outage plan.

Suggestions for improving this handbook should be sent to the maintenance engineer, who reviews them before each revision.